



Discussion Forums > All Course Discussions

Plotting the ellipsoid fit to the data assignment



Hanna, A A Learner

3 years ago

I don't understand why we used

```
ellipse_x=2*sqrt(eigvals(1))*cos(theta);
```

```
ellipse_y=2*sqrt(eigvals(2))*sin(theta);
```

instead of

```
ellipse_x=2*sqrt(eigvals(1))*eigvectors(1,1)*cos(theta) + 2*sqrt(eigvals(2))*eigvectors(1,2)*sin(theta);
```

```
ellipse_y=2*sqrt(eigvals(1))*eigvectors(2,1)*cos(theta) + 2*sqrt(eigvals(2))*eigvectors(2,2)*sin(theta);
```



Like 8



Reply 2



Follow this post More ...

2 Replies

Recent ▾

SR Sierra R. Learner

3 years ago

To understand better the reply to the original question, I plotted feature 1 ($D(:,1)$) vs feature 2 ($D(:,2)$), and used the following to plot the ellipse:

```
ellipse_x=2*sqrt(eigvals(1))*eigvectors(1,1)*cos(theta) + 2*sqrt(eigvals(2))*eigvectors(1,2)*sin(theta);
```

```
ellipse_y=2*sqrt(eigvals(1))*eigvectors(2,1)*cos(theta) + 2*sqrt(eigvals(2))*eigvectors(2,2)*sin(theta);
```

however, the ellipse is very far from the data points. Can you explain what is wrong here?



Like



Reply

More ...

JN Jack Noble Instructor

3 years ago

In the PCA linear vector space, the first PCA dimension *is* the x-axis and the second PCA dimension *is* the y-axis. Thus the major axis of the ellipse is aligned with the x-axis and the minor axis is aligned with the y-axis.

The second case is what you need to visualize the first two PCA dimensions in the first two dimensions of the inherent data coordinate system, where the x-axis corresponds to the value of feature 1 and the y-axis corresponds to the value of feature 2.



Like 8



Reply

More ...

